

Programming Project Phase 3 Rubric — RDT 3.0 over an Unreliable UDP Channel with Bit-Errors and Loss

Total: 100 points

Requirements

Phase 3(a): RDT 3.0 protocol implementation

- **R1:** Implement file transfer over UDP using **RDT 3.0** (Kurose/Ross, Section 3.4.1).
- **R2:** Use sequence numbers to detect duplicates.
- **R3:** Implement a checksum for error detection on DATA and ACK packets.
- **R4:** Implement ACK-based recovery (RDT 3.0 remains NAK-free, building on RDT 2.2).
- **R5:** Sender and receiver behavior must match the RDT 3.0 sender/receiver FSM semantics (note: RDT 3.0 receiver is the same as RDT 2.2 receiver).

Phase 3(b): New feature — countdown timer for packet loss

- **R6:** Enhance the sender to handle packet loss using a **countdown timer** (RDT 3.0).
- **R7:** The sender must retransmit the current packet when the timer expires without receiving a valid ACK.

Phase 3(c): Packetization and reassembly

- **R8:** Packetize the transfer file into a constant payload size per packet (example payload size: 1024 bytes).
- **R9:** Non-pipelined sending (send next packet only after the previous is acknowledged / retransmission logic completes).
- **R10:** Receiver must reassemble packets in order and deliver the full file to the application.
- **R11:** In the no-error mode, the output file must match the input file (byte-for-byte correctness preferred).

Phase 3(d): Required transfer scenarios

You must demonstrate all five scenarios below:

- **R12:** Option 1 — No loss and no bit-errors.
- **R13:** Option 2 — ACK packet bit-error injection at the sender-side receive path (and correct recovery).

- **R14:** Option 3 — DATA packet bit-error injection at the receiver-side receive path (and correct recovery).
- **R15:** Option 4 — ACK packet loss injection at the sender-side receive path (and correct recovery).
- **R16:** Option 5 — DATA packet loss injection at the receiver-side receive path (and correct recovery).

Phase 3(e): Performance evaluation and plots

- **R17:** Measure end-to-end completion time for transferring the same file under each option.
- **R18:** Produce plots of completion time versus loss/error rate from 0% to 95% in increments of 5%.
- **R19:** Each plotted point must be an average of at least 5 runs at that rate.
- **R20:** Remove (or disable) debug/print logging during timing experiments.

Plot requirements (what it should look like):

- **x-axis:** loss/error rate (%), using this list exactly:
 - `rates = [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]`
- **y-axis:** completion time in seconds (average of runs).

Recommended data structure for averaging:

- Store raw runs in a matrix (list of lists) per option:
 - `times_option1[r][k]` = completion time for rate index `r` and run index `k`
 - shape: `len(rates) x 5` (or more runs)
- Then compute:
 - `avg_option1[r] = mean(times_option1[r])`
 - Same for Options 2, 3, 4, and 5
- Final plot:
 - five lines on the same chart (Option 1, Option 2, Option 3, Option 4, Option 5), all against the same `rates` x-axis.

Deliverables and submission

- **R21:** Submit `DESIGN_DOCUMENT.pdf` or `DESIGN_DOCUMENT.md`.
 - If submitting on Blackboard, it must be PDF.
 - If submitting via GitHub only, Markdown is allowed.
- **R22:** Submit `README.pdf` or `README.md` with run instructions for all scenarios and how to reproduce plots.
- **R23:** Submit the plots (and include the raw timing data file(s) if you generated them).
- **R24:** Submit `contribution.txt` or `contribution.doc` (individual, confidential).

- **R25:** Repository setup:
 - Use one shared GitHub repo for the team (private is fine).
 - The repo must include all code, docs, and plot scripts.
 - The instructor must be invited to this repo and have access (verify access before submission).
 - **R26:** Demo video:
 - Submit a short YouTube (unlisted) or video link showing:
 - * Option 1 working end-to-end
 - * Option 2 working end-to-end
 - * Option 3 working end-to-end
 - * Option 4 working end-to-end
 - * Option 5 working end-to-end
 - * A quick look at the generated plot(s)
 - **R27:** Academic integrity: submissions may be inspected using plagiarism detection software.
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Grading breakdown

A) Design Document (required first) — 15 points

- **15:** Clear, complete design submitted before major coding. Includes:
 - Sender/receiver logic aligned with the RDT 3.0 FSM
 - Packet format (seq number, checksum, payload, ACK format)
 - Error injection approach for Options 2 and 3
 - Loss injection approach for Options 4 and 5
 - Countdown timer design and timeout behavior
 - Basic test plan and validation steps
- **10:** Mostly complete but missing one key element
- **5:** Thin/unclear or looks written after implementation
- **0:** Not submitted

B) RDT 3.0 Core Correctness — 30 points

- **30:** Correct behavior across normal operation, retransmissions, and timeout-based recovery; no protocol logic issues
- **24:** Mostly correct with minor issues (rare edge cases or minor logic inconsistencies)
- **16:** Partially correct (protocol breaks in at least one common case)
- **0–8:** Not working / cannot be demonstrated

C) Required Scenarios (Options 1–5) — 20 points

- **20:** All five options implemented and demonstrated correctly
- **16:** Four options correct
- **12:** Three options correct

- **8:** Two options correct
- **4:** One option correct
- **0:** No correct scenario handling

D) Performance Evaluation & Plots — 15 points

- **15:** Correct x/y axes, full rate range, averaging done correctly; plots are readable and labeled
- **10:** Minor issues (missing a few points, unclear labels, incomplete averaging evidence)
- **5:** Incomplete or incorrect methodology
- **0:** Missing

E) README & Reproducibility — 10 points

- **10:** Clear instructions to run all options, reproduce plots, and interpret outputs
- **6:** Instructions present but missing steps or ambiguous
- **3:** Minimal
- **0:** Missing

F) Repo, Contributions, and Demo Video — 10 points

- **10:** Repo access verified, meaningful commit history, contribution file submitted, and demo video complete
- **6:** Repo and contributions present but weak history and/or video is incomplete
- **3:** Minimal evidence of collaboration and/or unclear demo
- **0:** Missing repo access / missing contribution file / missing video

Grader notes template

Item	Max Pts	Earned	Notes / Deductions
A (Design Document)	15		
B (RDT 3.0 Core Correctness)	30		
C (Options 1–5)	20		
D (Performance & Plots)	15		
E (README & Reproducibility)	10		
F (Repo, Contributions, Demo Video)	10		
TOTAL	100		